

# WHERE DID THE COMMUNICATION BREAK?

Speaker / Anchor Script — Networking + SCADA Beginner Session

## Presentation goal

Run the session as an investigation rather than a lecture. The audience follows one real-world value — 78°C — from a temperature sensor to the SCADA screen. Your job is to ask questions, create pauses, and connect each networking concept back to that journey.

## Presenter rule

**Do not read the slides. Let the slide provide the visual; your voice provides the story, explanation and interaction.**

## SLIDE 1 — THE INCIDENT

SAY:

“Let's start with a situation.

Imagine we're sitting in a control room.

The bearing temperature sensor is working.

The PLC is running.

SCADA is also running.

And the sensor is telling us... 78 degrees Celsius.

But when the operator looks at SCADA... Communication Failure.

Now here's my question: Where exactly did the communication break?

Was it the sensor? The PLC? The network? The IP? A firewall? The protocol? Or SCADA itself?

By the end of this session, we're going to find out.”

**ANCHOR:**

**“So I want you to keep this 78°C in your mind. We're going to follow it.”**

**TRANSITION:**

**“But before we troubleshoot the failure, we need to understand one simple thing — what does communication between two devices actually require?”**

## SLIDE 2 — THE FOUNDATION

SAY:

“Let's forget networking terminology for just a minute.

Imagine I want to send something to you. What do I need to know?

First — who am I talking to?

Second — where is that device?

Third — which application or service am I actually trying to reach?

And finally — how are we going to communicate?

Networking has a lot of complicated words, but underneath those words we're really answering these four questions.”

### **INTERACTION:**

**“If I give you only an IP address, do you know everything required for communication?”**

Then:

“WHO — MAC. WHERE — IP. WHICH SERVICE — Port. HOW — Protocol.”

### **TRANSITION:**

**“So let's focus on the one everyone here has probably seen most often — IP.”**

## SLIDE 3 — IP ADDRESSING

SAY:

“Most of us have seen something like this: 10.10.10.20. We call it an IP address.

But the important question isn't just: what is the IP?

The important question is: what does this IP tell the network?

Our PLC is 10.10.10.20. Our SCADA is 10.10.10.100.

Now the network needs to determine: Are these two devices part of the same network?

That's where the subnet mask becomes important. It helps us determine what's local and what's outside our local network.

If the destination is local, we can communicate directly on that network. If it's somewhere else, we need a way out — the default gateway.”

### **ANALOGY:**

**“Think of your IP as your address. The subnet tells you who's in your neighborhood. And the gateway is basically the exit from your neighborhood.”**

### **TRANSITION:**

**“Okay, now we know where we're going. But who actually moves this traffic?”**

## SLIDE 4 — SWITCH / ROUTER / FIREWALL

SAY:

“This is where networking becomes much easier if we stop thinking about devices as boxes and start thinking about them as decision makers.

Switch: Where should this traffic go inside my local network?

Router: Which network should this traffic go to?

Firewall: Should this traffic be allowed at all?

So they're not doing the same job.

The switch decides locally. The router decides between networks. The firewall decides whether the traffic is allowed.

And if any one of these says ‘no’... our 78°C stops moving.”

### **TRANSITION:**

**“But we've been talking about traffic and data. What actually travels across this network?”**

## SLIDE 5 — ENCAPSULATION

SAY:

“Here's something interesting.

The network doesn't just take the number 78 and throw it onto the cable.

This is our actual application data. But to deliver it, different layers need to add information around it.

Transport information. IP information. Ethernet information. And eventually, it becomes something that can physically travel across the medium.

78°C is the payload. Everything around it helps deliver that payload.

Think about sending a gift. The gift is the actual data. The box, address label and shipping information help get the gift to the right destination. You don't change the gift just because you're transporting it.

And at the destination, those layers are processed in reverse until the application gets the original value.”

### **TRANSITION:**

**“Now let's bring this back to the system we actually care about — SCADA.”**

## SLIDE 6 — SCADA ARCHITECTURE

SAY:

“This is probably the most important connection I want everyone to take away from today's session.

SCADA is not the network.

SCADA uses the network.

Think about our temperature value. The sensor measures it. The PLC receives and holds that value. Some communication protocol carries that information. The network transports it. The SCADA driver receives and interprets it. And finally, the operator sees a tag on the HMI.

So when SCADA says communication failure... it doesn't automatically mean SCADA is the problem.

SCADA is sitting at the end of a chain. Anything before it can break that chain.”

### **TRANSITION:**

**“But there's still one missing piece.”**

## SLIDE 7 — PROTOCOLS

SAY:

“We've established the road. We've established the destination. But devices still need to agree on how they're going to communicate.

That's where protocols come in.

IP tells us: Where is the device?

The port tells us: Which service am I trying to reach?

And the protocol tells us: How are the applications going to communicate?

In an industrial environment, you might encounter things like Modbus TCP, OPC UA, IEC 104.

The exact protocol can change, but the underlying network still has to work.

A perfect protocol cannot save a broken network.”

### **TRANSITION:**

**“So now we've learned all the individual pieces. Let's stop talking about the pieces and follow our actual data.”**

## SLIDE 8 — THE INVESTIGATION

SAY:

“Let's follow the 78 degrees.

Step one — the sensor measures 78°C.

The PLC receives and holds that value.

The industrial protocol defines how the application communicates.

The networking stack prepares the data for transport.

Now the infrastructure has to forward it, and the firewall has to allow it.

Then the SCADA driver receives it, maps it to the correct tag, and finally the operator sees 78°C.

Now tell me...

At which point could our 78°C disappear?”

Let the audience answer. Do not immediately correct them.

“Exactly. Potentially at multiple points.

And that's why troubleshooting without a structured approach becomes guesswork.”

### **TRANSITION:**

**“So let's go back to our original incident.”**

## SLIDE 9 — SOLVE THE INCIDENT

SAY:

“Remember our incident? PLC is running. SCADA is running. But we don't see the data.

PLC? Running. Good.

Switch link? Up. Good.

Let's ping the PLC. Success.

So... is everything working?”

Pause and let the room respond.

“No.

Because ping proves basic reachability. It doesn't prove that the actual application service is reachable.

And here we have... TCP connection refused.

So where would you investigate next?”

Let the audience choose: port, firewall, service, protocol or SCADA driver.

Then:

“Based on the evidence we've been given, the basic network path is reachable. But the required TCP service isn't accepting the connection. So our investigation moves upward — toward the service, port/firewall, protocol or application side.

The mistake is to say: ‘SCADA is not working.’ The better question is: ‘Which layer is not working?’”

## SLIDE 10 — TAKEAWAY

SAY:

“So, if you remember only one thing from today's session...

Don't try to memorize networking.

Follow the data.

When SCADA says communication failure, don't immediately blame SCADA. Don't immediately blame the network. Don't immediately change configuration.

Ask: Where is my data right now?

Is the physical connection working?

Can the devices communicate?

Is the IP path correct?

Is the required service reachable?

Is the protocol working?

Is the SCADA driver receiving the data?

Don't guess. Follow the data path.

And that's how we find where the communication broke.

Thank you. Questions?”